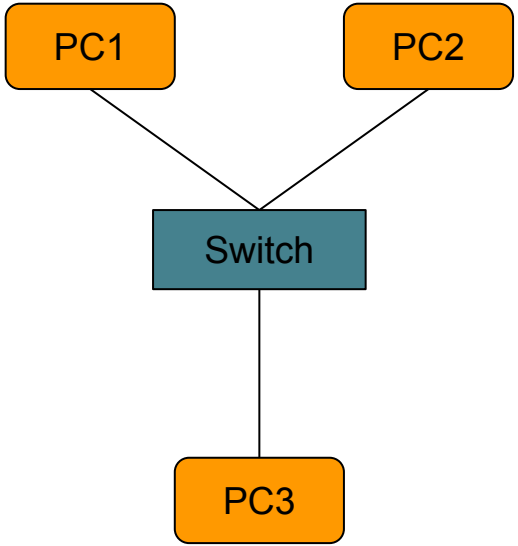
Tópicos Avançados em Novas Tecnologias

Primeiros passos com docker

Prof. Diego Cardoso Borda Castro

Network

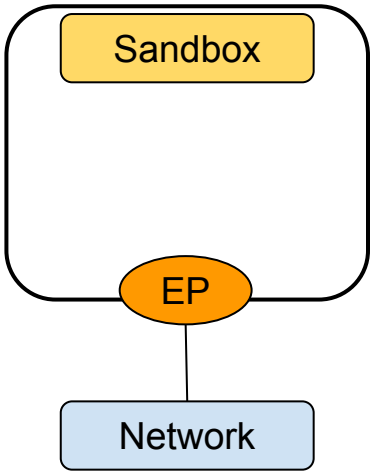
- Rede tradicional
  - Interfaces de rede físicas e virtuais
  - IP, MAC e tabela de rotas
  - Switch conecta vários hosts
  - Todos compartilham o mesmo namespace de rede
  - No Linux normal, todos os processos usam a mesma rede



Network

- Rede de containers
  - Cada processo pode ter sua própria rede
  - Interfaces isoladas
  - IP isolado
  - Tabela de rotas isolada

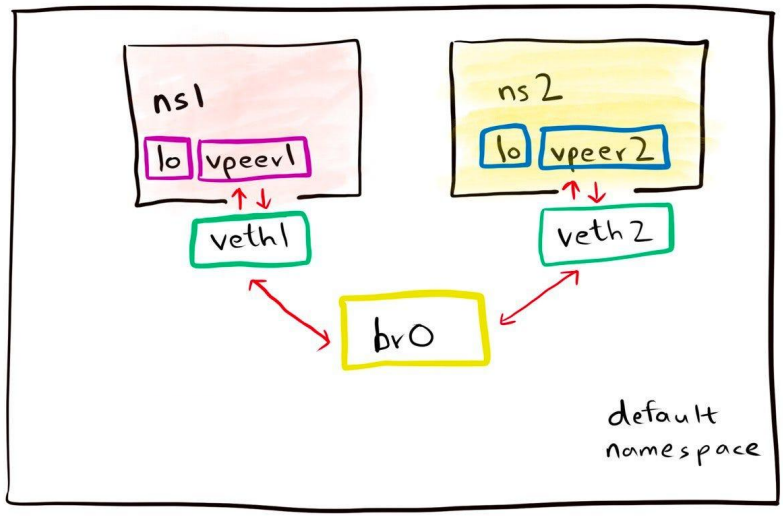
```
root@diego-XPS-13-9300: /home/diego
root@diego-XPS-13-9300:/home/diego# ip netns add ns1
root@diego-XPS-13-9300:/home/diego# ip netns add ns2
root@diego-XPS-13-9300:/home/diego# ip netns list
ns2
ns1
root@diego-XPS-13-9300:/home/diego#
```



Network

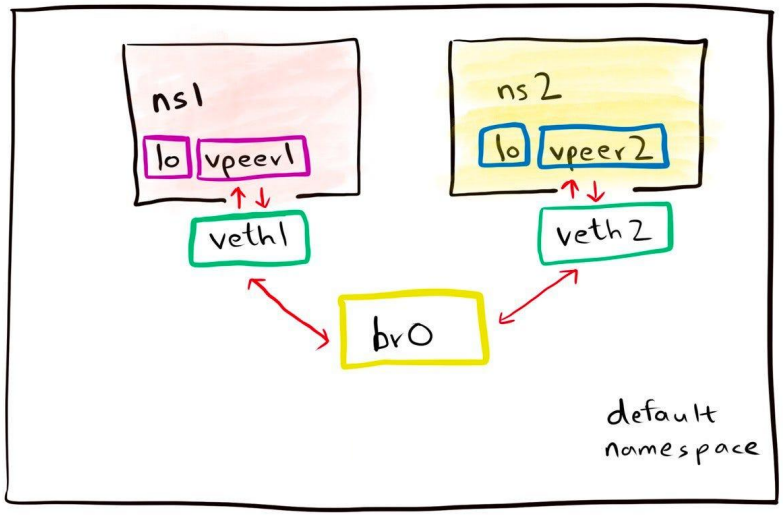
- Bridge Linux
  - Funciona como switch
  - Conecta várias interfaces
  - Permite comunicação entre namespaces

```
root@diego-XPS-13-9300:/home/diego# ip link add br0 type bridge
root@diego-XPS-13-9300:/home/diego# ip link set br0 up
root@diego-XPS-13-9300:/home/diego# ip link show br0
37: br0: <NO-CARRIER,BROADCAST,MULTICAST,UP> mtu 1500 qdisc noqueue st
len 1000
    link/ether 5a:83:95:13:bb:c6 brd ff:ff:ff:ff:ff:ff
root@diego-XPS-13-9300:/home/diego#
```



Network

- Virtual Ethernet (veth)
  - Par de interfaces virtuais
  - Como um cabo virtual
  - Liga dois namespaces

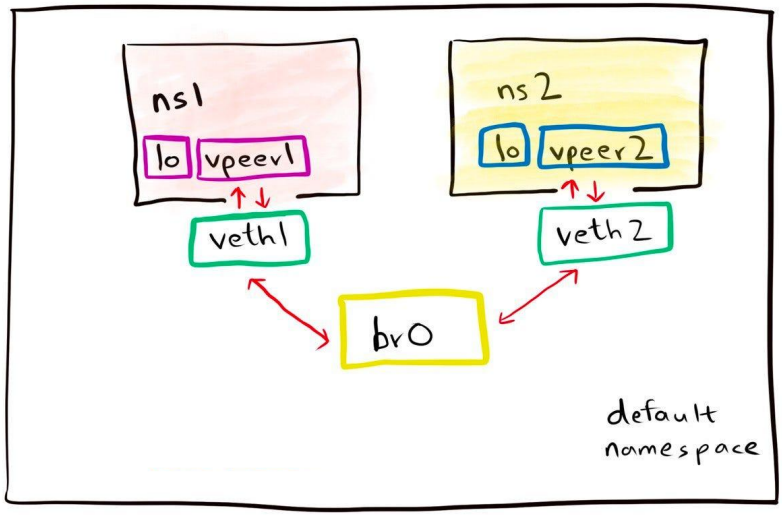



```
root@diego-XPS-13-9300: /home/diego
root@diego-XPS-13-9300:/home/diego# ip link add veth1 type veth peer name vpeer1
root@diego-XPS-13-9300:/home/diego# ip link add veth2 type veth peer name vpeer2
root@diego-XPS-13-9300:/home/diego# ip link set vpeer1 netns ns1
root@diego-XPS-13-9300:/home/diego# ip link set vpeer2 netns ns2
root@diego-XPS-13-9300:/home/diego#
```

Network

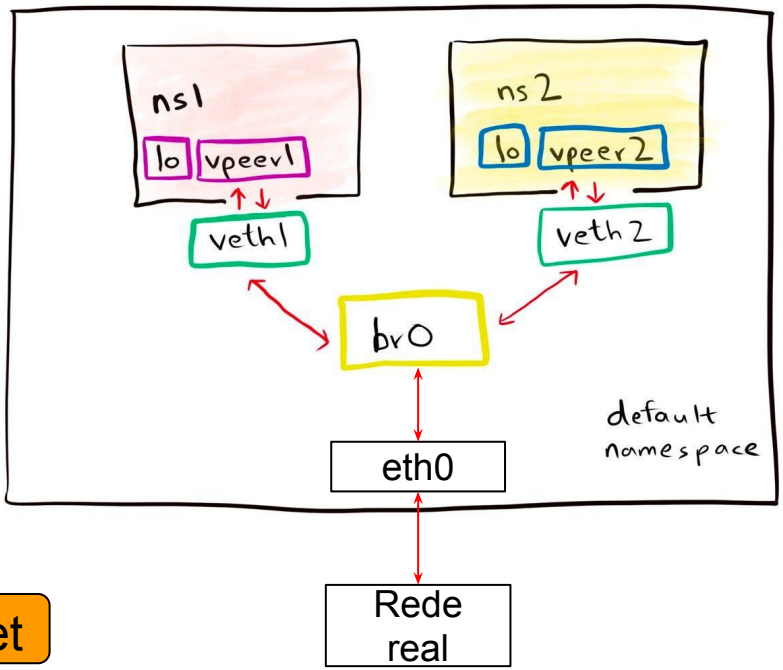
```
root@diego-XPS-13-9300:/home/diego# ip link set veth1 master br0
root@diego-XPS-13-9300:/home/diego# ip link set veth2 master br0
```

```
root@diego-XPS-13-9300: /home/diego
root@diego-XPS-13-9300:/home/diego# ip netns exec ns1 ip link set lo up
root@diego-XPS-13-9300:/home/diego# ip netns exec ns1 ip link set vpeer1 up
root@diego-XPS-13-9300:/home/diego# ip netns exec ns1 ip addr add 10.0.0.1/24 dev vpeer1
root@diego-XPS-13-9300:/home/diego# ip netns exec ns2 ip link set lo up
root@diego-XPS-13-9300:/home/diego# ip netns exec ns2 ip link set vpeer2 up
root@diego-XPS-13-9300:/home/diego# ip netns exec ns2 ip addr add 10.0.0.2/24 dev vpeer2
root@diego-XPS-13-9300:/home/diego# ip addr add 10.0.0.254/24 dev br0
root@diego-XPS-13-9300:/home/diego# ip link set br0 up
root@diego-XPS-13-9300:/home/diego# ip netns exec ns1 ping 10.0.0.2
PING 10.0.0.2 (10.0.0.2) 56(84) bytes of data.
64 bytes from 10.0.0.2: icmp_seq=1 ttl=64 time=0.108 ms
64 bytes from 10.0.0.2: icmp_seq=2 ttl=64 time=0.059 ms
64 bytes from 10.0.0.2: icmp_seq=3 ttl=64 time=0.069 ms
64 bytes from 10.0.0.2: icmp_seq=4 ttl=64 time=0.071 ms
64 bytes from 10.0.0.2: icmp_seq=5 ttl=64 time=0.064 ms
64 bytes from 10.0.0.2: icmp_seq=6 ttl=64 time=0.063 ms
64 bytes from 10.0.0.2: icmp_seq=7 ttl=64 time=0.067 ms
64 bytes from 10.0.0.2: icmp_seq=8 ttl=64 time=0.092 ms
64 bytes from 10.0.0.2: icmp_seq=9 ttl=64 time=0.062 ms
64 bytes from 10.0.0.2: icmp_seq=10 ttl=64 time=0.058 ms
```



Network

- Ethernet
  - eth0 → interface física
  - Acesso a internet
    - Saída pra rua




```
~
ip link set wlp0s20f3 master br0
```

ns1 → veth → br0 → eth0 → rede

container → veth → docker0 → eth0 → internet

Network

- Listagem das interfaces de rede do linux
  - Redes virtuais
  - Redes reais
  - Redes docker



Network

- Docker network
  - Usa namespaces
  - Usa veth
  - Usa bridge

```
root@diego-XPS-13-9300: /home/diego
root@diego-XPS-13-9300:/home/diego# docker network ls
NETWORK ID     NAME                                                DRIVER    SCOPE
6b817a2ed5de   bridge                                              bridge    local
54c42d2ccd13   direct_default                                      bridge    local
2f0501812150   exemplo1_default                                    bridge    local
6f907e69f612   exemplo2_default                                    bridge    local
dd33ab377122   exemplo4_default                                    bridge    local
751b97122def   exemplobase_default                                 bridge    local
354f6f21a5d2   exemplocomcamadas_default                           bridge    local
0457b7e4cf26   host                                                host      local
60608122dee2   jaeger_default                                      bridge    local
f3cb40a1f541   none                                                null      local
53bf89e08782   postgresql_postgres-compose-network                 bridge    local
a02e73ac5cb7   rabbitmq-walkthrough-v1-master_walkthrough-net      bridge    local
root@diego-XPS-13-9300:/home/diego#
```

Docker network inspect ID

Network

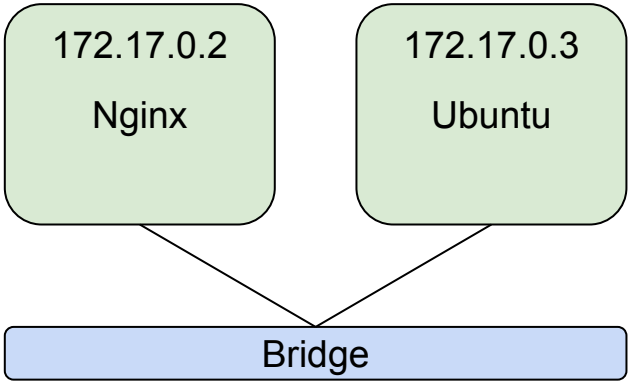
```
root@diego-XPS-13-9300: /home/diego
root@diego-XPS-13-9300:/home/diego# docker network inspect 6b817a2ed5de
[
    {
        "Name": "bridge",
        "Id": "6b817a2ed5deba021b4765c608c87f41848be70c28da7f15e6a426a2df0980b0",
        "Created": "2026-03-09T12:12:13.712441187-03:00",
        "Scope": "local",
        "Driver": "bridge",
        "EnableIPv4": true,
        "EnableIPv6": false,
        "IPAM": {
            "Driver": "default",
            "Options": null,
            "Config": [
                {
                    "Subnet": "172.17.0.0/16",
                    "Gateway": "172.17.0.1"
                }
            ]
        },
```

Network


"80/tcp": null
},
"HairpinMode": false,
"LinkLocalIPv6Address": "",
"LinkLocalIPv6PrefixLen": 0,
"SecondaryIPAddresses": null,
"SecondaryIPv6Addresses": null,
"EndpointID": "50168c2a9e6278572c1eb99c1cc4b6
"Gateway": "172.17.0.1",
"GlobalIPv6Address": "",
"GlobalIPv6PrefixLen": 0,
"IPAddress": "172.17.0.2",
"IPPrefixLen": 16,
"IPv6Gateway": "",
"MacAddress": "7e:4d:c5:65:26:f0",
"Networks": {
"bridge": {
"IPAMConfig": null,
"Links": null,
"Aliases": null,
"MacAddress": "7e:4d:c5:65:26:f0",
"DriverOpts": null,
"GwPriority": 0,
Nginx


"SecondaryIPAddresses": null,
"SecondaryIPv6Addresses": null,
"EndpointID": "",
"Gateway": "",
"GlobalIPv6Address": "",
"GlobalIPv6PrefixLen": 0,
"IPAddress": "",
"IPPrefixLen": 0,
"IPv6Gateway": "",
"MacAddress": "",
"Networks": {
"bridge": {
"IPAMConfig": null,
"Links": null,
"Aliases": null,
"MacAddress": "",
"DriverOpts": null,
"GwPriority": 0,
"NetworkID": "6b817a2ed5deba021b4765c608c8
"EndpointID": "",
"Gateway": "",
"IPAddress": "",
"IPPrefixLen": 0,
"IPv6Gateway": "",
"GlobalIPv6Address": "",
"GlobalIPv6PrefixLen": 0,
Ubuntu

Network



172.17.0.0/16

ping 172.17.0.2 ✅
ping nginx ❌

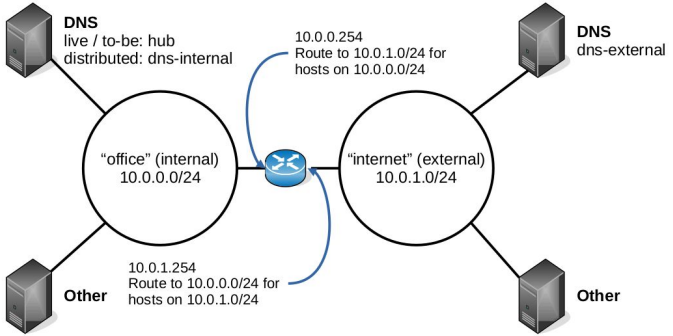
```
root@e2a5669706f3: /
root@e2a5669706f3:/# curl http://172.17.0.2
<!DOCTYPE html>
<html>
<head>
<title>Welcome to nginx!</title>
<style>
html { color-scheme: light dark; }
body { width: 35em; margin: 0 auto;
font-family: Tahoma, Verdana, Arial, sans-serif; }
</style>
</head>
<body>
<h1>Welcome to nginx!</h1>
<p>If you see this page, nginx is successfully installed and working.
Further configuration is required for the web server, reverse proxy,
API gateway, load balancer, content cache, or other features.</p>

<p>For online documentation and support please refer to
<a href="https://nginx.org/">nginx.org</a>.<br/>
To engage with the community please visit
<a href="https://community.nginx.org/">community.nginx.org</a>.<br/>
For enterprise grade support, professional services, additional
security features and capabilities please refer to
<a href="https://f5.com/nginx">f5.com/nginx</a>.</p>
```

Não é uma boa prática

Network

- DNS e resolução de nome
  - Boa prática
  - Não marretar o IP



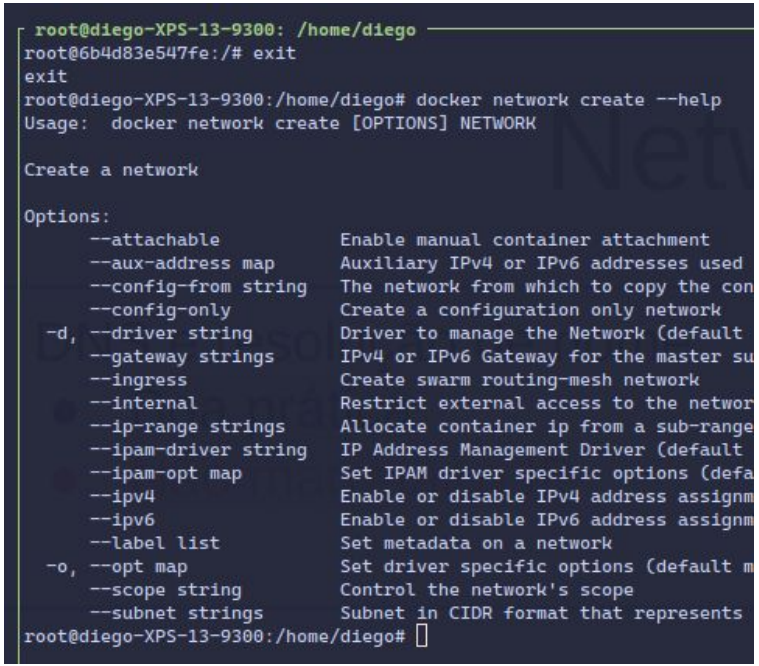
Docker network connect aula_docker nginx

```
root@diego-XPS-13-9300: /home/diego
root@diego-XPS-13-9300:/home/diego# docker network create aula_docker
2d081df106498325c314beed57ecd06b137e20a948e669011546cac39fcde673
root@diego-XPS-13-9300:/home/diego# docker network ls
NETWORK ID     NAME                                      DRIVER
2d081df10649   aula_docker                               bridge
6b817a2ed5de   bridge                                    bridge
```

```
root@6b4d83e547fe: /
root@6b4d83e547fe:/# curl http://nginx
<!DOCTYPE html>
<html>
<head>
<title>Welcome to nginx!</title>
<style>
html { color-scheme: light dark; }
body { width: 35em; margin: 0 auto;
```

Network

- DNS e resolução de nome
  - docker network create --subnet=10.0.0.0/16 --gateway=10.0.0.1 rede_nova
  - docker container run -it ubuntu /bin/base



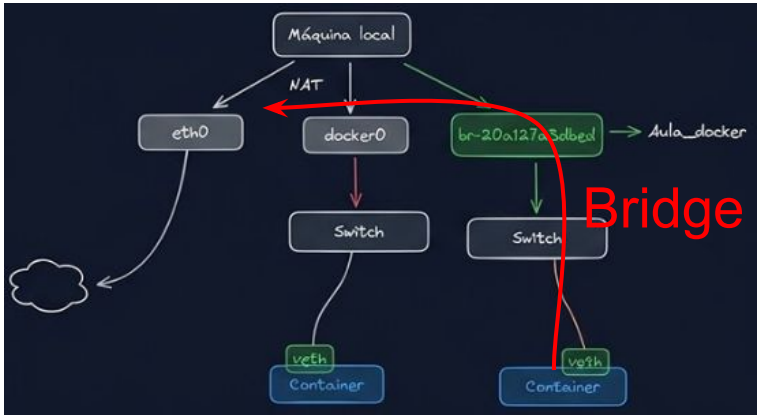
Network

- Mudar o padrão da rede bridge
  - daemon.json
  - https://docs.docker.com/engine/network/drivers/bridge/#use-the-default-bridge-network



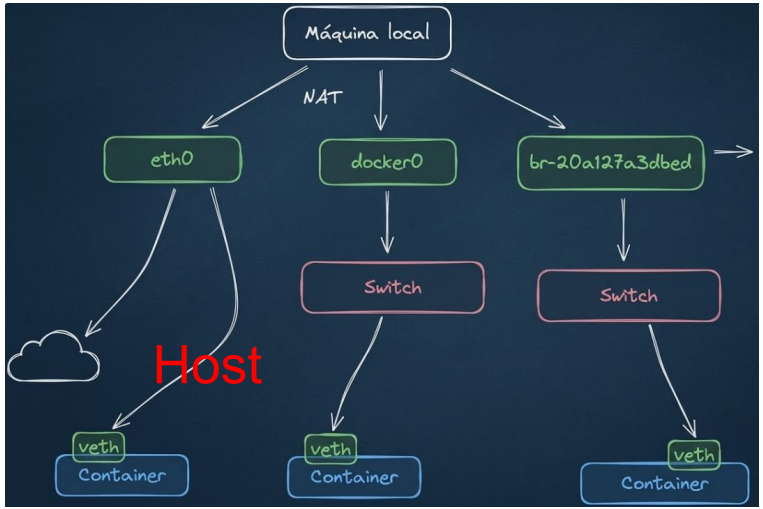
Network

- Redes
  - Bridge
    - Comunicação entre containers
    - Port Bind
  - Host
    - Remove isolamento entre o container e sua máquina
  - Overlay
    - Comunica vários docker hosts
  - Macvlan
    - Simula um endereço mac



Network

- Redes
  - Bridge
    - Comunicação entre containers
    - Port Bind
  - Host
    - Remove isolamento entre o container e sua máquina
    - Usa as portas da própria máquina
  - Overlay
    - Comunica vários docker hosts
  - Macvlan
    - Simula um endereço mac

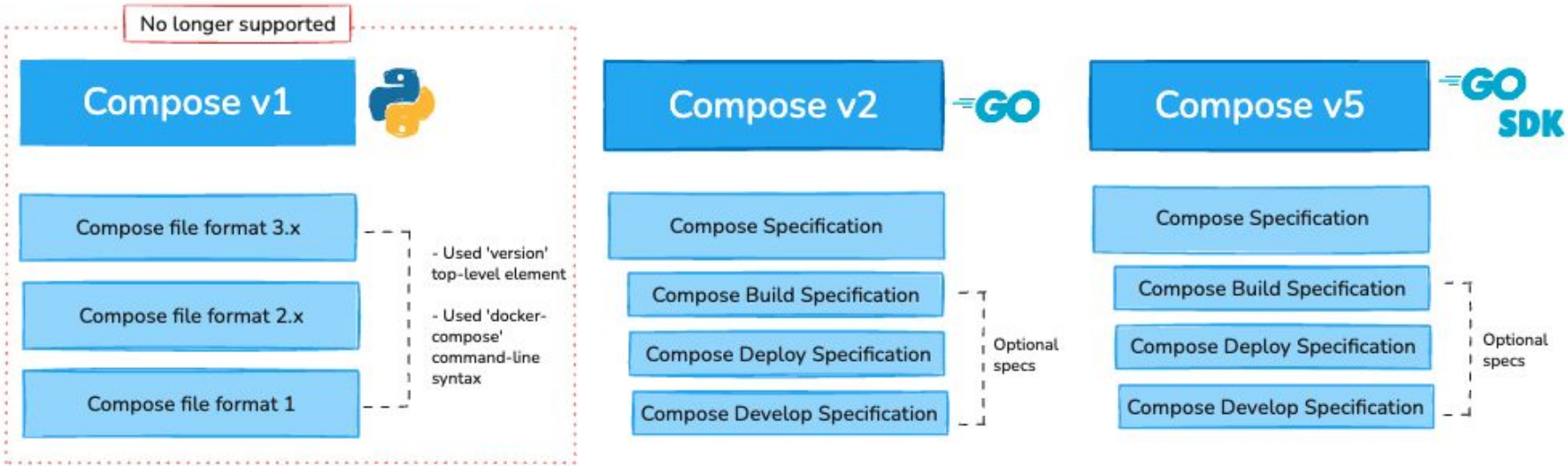


Docker compose

- Arquivo yaml com uma receita para criação de toda sua infraestrutura
  - Container
  - Volumes
  - Networks
  - …

- Dockerfile cria imagem
- Docker compose cria containers

Docker compose

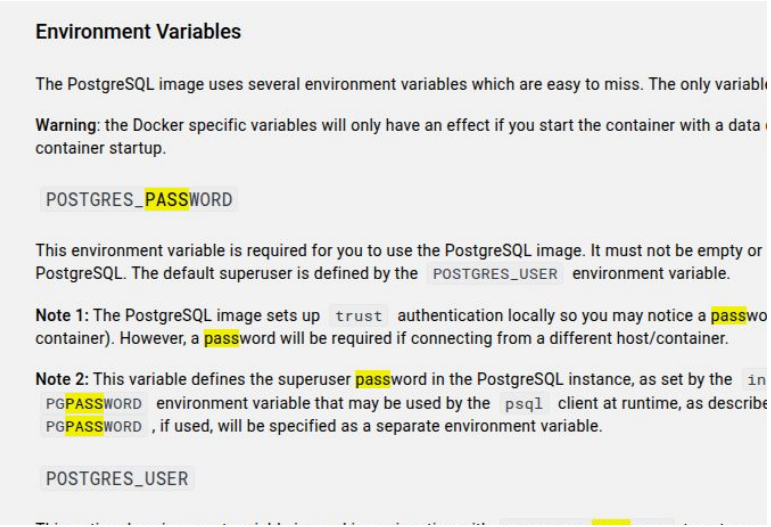


Docker compose

- v1
  - docker-compose
- v2
  - docker compose
- Especificação da stack de containers
  - https://www.compose-spec.io/
    - Você pode usar o mesmo compose em:
      - Kubernetes (via conversão)
      - Cloud (AWS, Azure, GCP)
      - CI/CD

Docker compose

- Variáveis de ambiente



Environment Variables

The PostgreSQL image uses several environment variables which are easy to miss. The only variabl

Warning: the Docker specific variables will only have an effect if you start the container with a data
container startup.

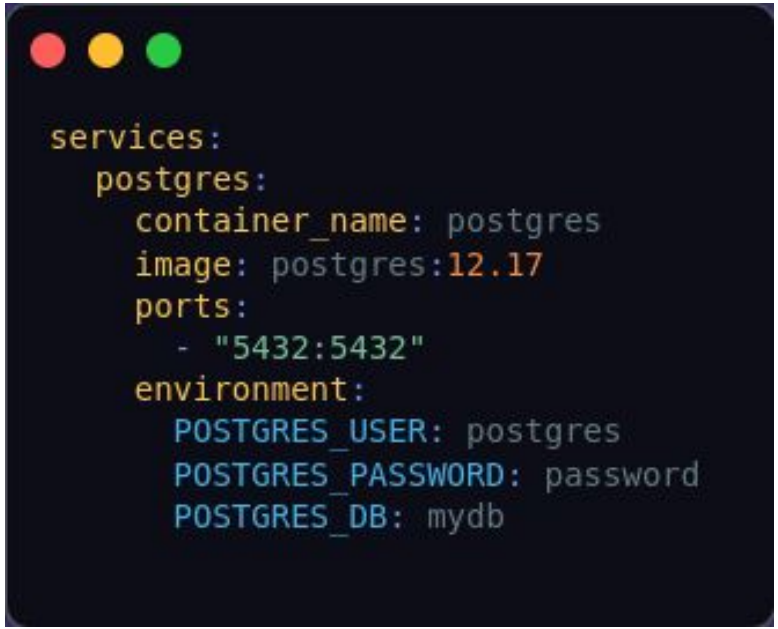
`POSTGRES_PASSWORD`

This environment variable is required for you to use the PostgreSQL image. It must not be empty or
PostgreSQL. The default superuser is defined by the `POSTGRES_USER` environment variable.

Note 1: The PostgreSQL image sets up `trust` authentication locally so you may notice a passwo
container). However, a password will be required if connecting from a different host/container.

Note 2: This variable defines the superuser password in the PostgreSQL instance, as set by the `in`
`PGPASSWORD` environment variable that may be used by the `psql` client at runtime, as describe
`PGPASSWORD`, if used, will be specified as a separate environment variable.

`POSTGRES_USER`



```
services:
  postgres:
    container_name: postgres
    image: postgres:12.17
    ports:
      - "5432:5432"
    environment:
      POSTGRES_USER: postgres
      POSTGRES_PASSWORD: password
      POSTGRES_DB: mydb
```

Docker compose

- docker compose run -d -p 8080:80 nginx:latest
- Criar arquivo docker-compose.yaml / compose.yaml
  - docker compose -f compose.yaml up -d
  - docker compose down
  - docker compose stop
  - docker compose start



```yaml
# version: '3.8'

services:
  nginx:
    container_name: nginx
    image: nginx:latest
    ports:
      - "8080:80"
    commands:
      - "nginx -g 'daemon off;'"
```

Docker compose

```
docker compose up -d
[+] Running 15/15
 ✔ postgres Pulled                                                                                                10.2s
WARN[0010] Found orphan containers ([nginx]) for this project. If you removed or renamed this service in your compose file, you can run this
 command with the --remove-orphans flag to clean it up.
[+] Running 1/1
 ✔ Container postgres  Started                                                                                     0.2s
```

- docker compose up -d --remove-orphans --build

Docker compose

Bind Mount

```yaml
services:
  postgres:
    container_name: postgres
    image: postgres:12.17
    ports:
      - "5432:5432"
    environment:
      - POSTGRES_USER=postgres
      - POSTGRES_PASSWORD=postgres
      - POSTGRES_DB=postgres
    volumes:
      - ./postgres_data:/var/lib/postgresql/data
```

Nomeado

```yaml
services:
  postgres:
    container_name: postgres
    image: postgres:12.17
    ports:
      - "5432:5432"
    environment:
      - POSTGRES_USER=postgres
      - POSTGRES_PASSWORD=postgres
      - POSTGRES_DB=postgres
    volumes:
      - postgres_data:/var/lib/postgresql/data

volumes:
  postgres_data:
    name: postgres_data
    external: true
```

Docker compose

```yaml
services:
  postgres:
    container_name: postgres
    image: postgres:12.17
    ports:
      - "5432:5432"
    environment:
      - POSTGRES_USER=postgres
      - POSTGRES_PASSWORD=postgres
      - POSTGRES_DB=postgres
    volumes:
      - postgres_data:/var/lib/postgresql/data
    networks:
      - postgres_network
```

```yaml
  pgadmin:
    container_name: pgadmin
    image: dpage/pgadmin4:latest
    ports:
      - "8080:80"
    environment:
      - PGADMIN_DEFAULT_EMAIL=admin@admin.com
      - PGADMIN_DEFAULT_PASSWORD=admin
    depends_on:
      - postgres
    networks:
      - postgres_network

volumes:
  postgres_data:
    name: postgres_data
    external: true

networks:
  postgres_network:
    driver: bridge
```

Docker compose

APP

```
const dbHost = process.env.DB_HOST || "localhost";

console.log("DB HOST:", dbHost);
```

Dockerfile

```
FROM node:20
WORKDIR /app
COPY package*.json ./
RUN npm install
COPY . .
ENV DB_HOST=localhost
CMD ["node", "app.js"]
```

Compose

```
services:
  app:
    build: .
    environment:
      - DB_HOST=postgres

  postgres:
    image: postgres
```

Docker compose

- extra_host
  - Entradas no /etc/hosts dentro do container

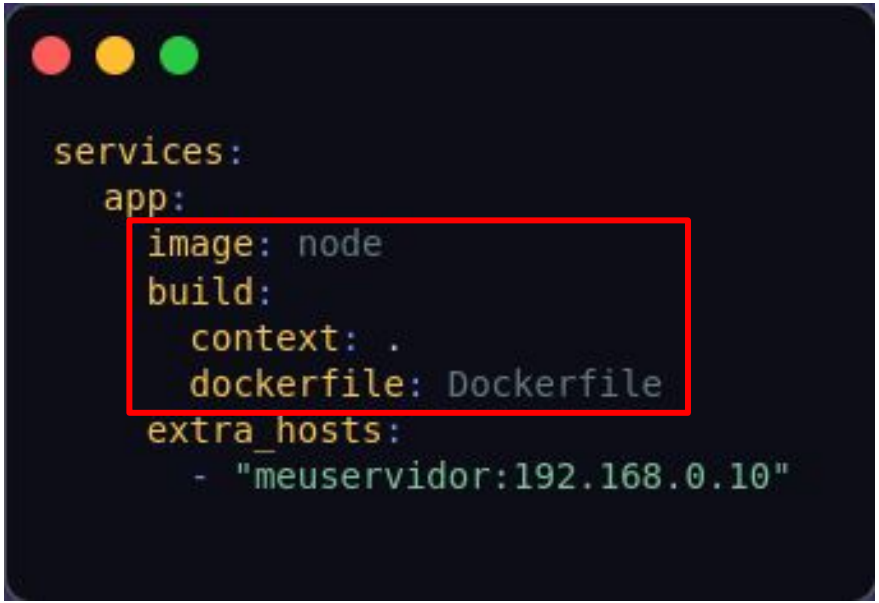
```yaml
services:
  app:
    image: node
    extra_hosts:
      - "meuservidor:192.168.0.10"
```

host.docker.internal:host-gateway

```yaml
services:
  postgres:
    container_name: postgres
    image: postgres:12.17
    ports:
      - "5432:5432"
    environment:
      - POSTGRES_USER=postgres
      - POSTGRES_PASSWORD=postgres
      - POSTGRES_DB=postgres
    volumes:
      - postgres_data:/var/lib/postgresql/data
    network_mode: host
```

Docker compose

- Construindo a imagem junto com o compose
- Atributo build
- docker compose up -d --build
- docker compose build
- docker compose push



Docker compose

- Variáveis de ambiente
  - POSTGRES_VERSION=17 docker compose up -d
  - docker compose config
  - docker compose up -d --env-file=arquivo.env

```yaml
services:
  postgres:
    container_name: postgres
    image: postgres:${POSTGRES_VERSION:-latest}
    ports:
      - "5432:5432"
    environment:
      - POSTGRES_USER=postgres
      - POSTGRES_PASSWORD=postgres
      - POSTGRES_DB=postgres
    volumes:
      - postgres_data:/var/lib/postgresql/data
    network_mode: host
```

Docker compose

- Múltiplos compose
  - Extends
  - Merge
  - Include

```yaml
services:
  postgres:
    extends:
      file: postgres.compose.yaml
      service: postgres
    networks:
      - postgres_network
```

```yaml
services:
  postgres:
    container_name: postgres
    image: postgres:${POSTGRES_VERSION:-latest}
    ports:
      - "5432:5432"
    environment:
      - POSTGRES_USER=postgres
      - POSTGRES_PASSWORD=postgres
      - POSTGRES_DB=postgres
    volumes:
      - postgres_data:/var/lib/postgresql/data
    network_mode: host
```

Docker compose

- Múltiplos compose
  - Extends
  - Merge
    - -f arquivo1
      -f arquivo 2
  - Include

```yaml
services:
  postgres:
    image: postgres:17
```

compose.override.yaml

```yaml
services:
  postgres:
    container_name: postgres
    image: postgres:${POSTGRES_VERSION:-latest}
    ports:
      - "5432:5432"
    environment:
      - POSTGRES_USER=postgres
      - POSTGRES_PASSWORD=postgres
      - POSTGRES_DB=postgres
    volumes:
      - postgres_data:/var/lib/postgresql/data
    network_mode: host
```

compose.yaml

Docker compose

- Múltiplos compose
  - Extends
  - Merge
  - Include

```yaml
services:
include:
  - ./arquivo.compose.yaml
```



```yaml
services:
  postgres:
    container_name: postgres
    image: postgres:${POSTGRES_VERSION:-latest}
    ports:
      - "5432:5432"
    environment:
      - POSTGRES_USER=postgres
      - POSTGRES_PASSWORD=postgres
      - POSTGRES_DB=postgres
    volumes:
      - postgres_data:/var/lib/postgresql/data
    network_mode: host
```

Docker compose

- Profile
  - Perfis de execução do compose
  - docker compose -- profile dev up -d

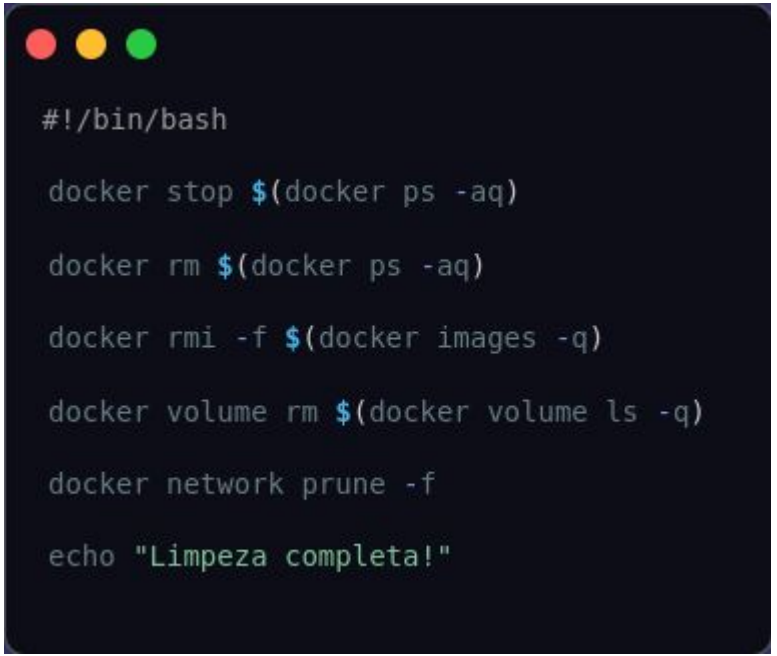
```yaml
services:
  postgres:
    container_name: postgres
    image: postgres:${POSTGRES_VERSION:-latest}
    ports:
      - "5432:5432"
    profiles:
      - dev
      - homolog
    environment:
      - POSTGRES_USER=postgres
      - POSTGRES_PASSWORD=postgres
      - POSTGRES_DB=postgres
    volumes:
      - postgres_data:/var/lib/postgresql/data
    network_mode: host
```

Últimas Dicas

- docker compose ls
- docker compose logs
- docker compose logs service name
- docker compose exec service name
- docker compose up
- docker compose start
- docker compose stop
- docker compose down
- docker compose ps
- docker compose build
- docker compose build app
- docker compose restart
- docker compose restart app
- docker compose pull
- docker compose rm app
- docker compose kill app

Últimas Dicas

- drop_hard.sh
  - Criar arquivo para limpeza de tudo referente ao docker na sua máquina
- docker system prune



```
#!/bin/bash

docker stop $(docker ps -aq)

docker rm $(docker ps -aq)

docker rmi -f $(docker images -q)

docker volume rm $(docker volume ls -q)

docker network prune -f

echo "Limpeza completa!"
```